

SMART ROBOT WITH ADVANCED LINE FOLLOWING CAPABILITIES AND INTEGRATED SENSORS



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INTERNAL GUIDE:TANVEER SULTANA

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Introduction:



- **Smart Car System: Redefining driving experience through cutting-edge technologies.**
- **Goal: Introduce intelligent features and seamless connectivity for safer and more efficient driving.**
- **Global Accessibility: Designed for worldwide users, envisioning intelligent systems prioritizing safety and efficiency**

OBJECTIVE:

- **Upgrade black line follower robots with IoT for smarter, precise, and adaptable functionality.**
- **Features: Integrate computer vision and sensor fusion to overcome navigation limitations.**
- **Focus: Develop a cost-effective circuit design for real-time line tracking and expanded applications.**
- **Vision: Enhance robots' usefulness in various industries and homes, ushering in a new era of IoT-enabled excellence.**



RESEARCH PAPERS 2011:

A Line Follower Robot from design to Implementation: Technical issues and problems

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***Abstract*— the Line follower robot is a mobile machine that can detect and follow the line drawn on the floor. Generally, the path is predefined and can be either visible like a black line on a white surface with a high contrasted color or it can be invisible like a magnetic field.**

Therefore, this kind of Robot should sense the line with its Infrared Ray (IR) sensors that installed under the robot. After that, the data is transmitted to the processor by specific transition buses. Hence, the processor is going to decide the proper commands and then it sends them to the driver and thus the path will be followed by the line follower robot.

TABAR is a line follower robot designed and tested in order to attend at Tabrize line follower robots competition. But it encounter with some technical and mechanical problems. In this Paper, we have illustrated the process of design, implementation and testing TABAR, a small line follower robot designed for the line follower robots competition. The technical and mechanical issues and problems also have investigated.

Keywords—Line Follower; Problems and solutions; Circuit; Actuator; Programming.

I. INTRODUCTION

Generally, the line follower robot is one of the self-operating mobile machines that follows a line drawn on the floor. The path can be a visible black line on a white surface (reverse). The basic operations of the line follower are as

This kind of robot can be used for military purposes, delivery services, transportation systems, blind assistive applications. Moreover, there are many annual line follower robots competitions organized by universities or industries around the world. They usually ask robotic teams for building a small robot with specific dimensions and weight according to the competition rules.

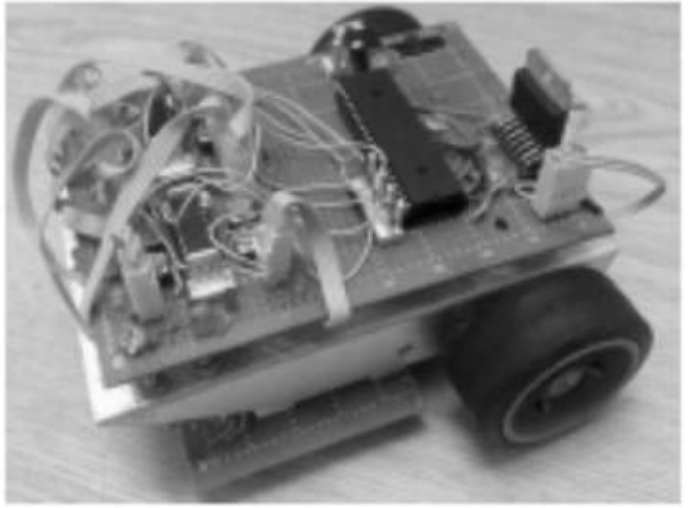


Figure 1. The Designed Line follower robot (TABAR)

In order to attend at Tabriz line follower robots competition, Tabari Institute of Babol had functional



APPLICATIONS:

- Industrial Automation and Material Handling
- Warehouse Logistics and Inventory Management
- Smart Home Cleaning Automation
- Last-Mile Delivery Services
- Surveillance and Security Operations
- Education and Research in Robotics
- Outdoor Agriculture Monitoring
- Public Spaces Navigation and Assistance
- Healthcare Supply Transport
- Smart City Infrastructure Applications



The existing system

- Deficiency: Lack of object detection severely hampers robot efficiency and poses risks.
- Limitations: The inability to recognize and avoid obstacles impedes practical applications.
- Importance: Addressing this deficiency is crucial for enhancing safety, adaptability, and efficiency.

Proposed system

- Integration: Seamlessly integrate object detection to address prior deficiencies.
- Transformation: Propel robots into a new league of precision and adaptability.
- Technologies: Infuse computer vision, sensor fusion, and IoT for enhanced decision-making.
- Standards: Set new benchmarks for efficiency and safety, opening avenues in logistics and surveillance.



ADVANTAGES OF THE PROPOSED SYSTEM:

- Enhanced Precision
- Real-time Adaptability
- Improved Safety
- Global Accessibility
- Cost-Effective Circuit Design
- Versatility in Applications
- Efficient Decision-Making
- User-Friendly Interface
- Expanded Potential for Industries
- Contribution to IoT Advancements



STRUCTURE OF ROBOT:

This robot can be divided into several parts:

- Sensors
- ADC (Analog to Digital Converter) and sensor circuit
- Processor Volume 1 978-1-4244-5586-7/10/ 2010 IEEE C 6
- Driver
- Actuators (Motors and wheels)
- Chassis and body structure

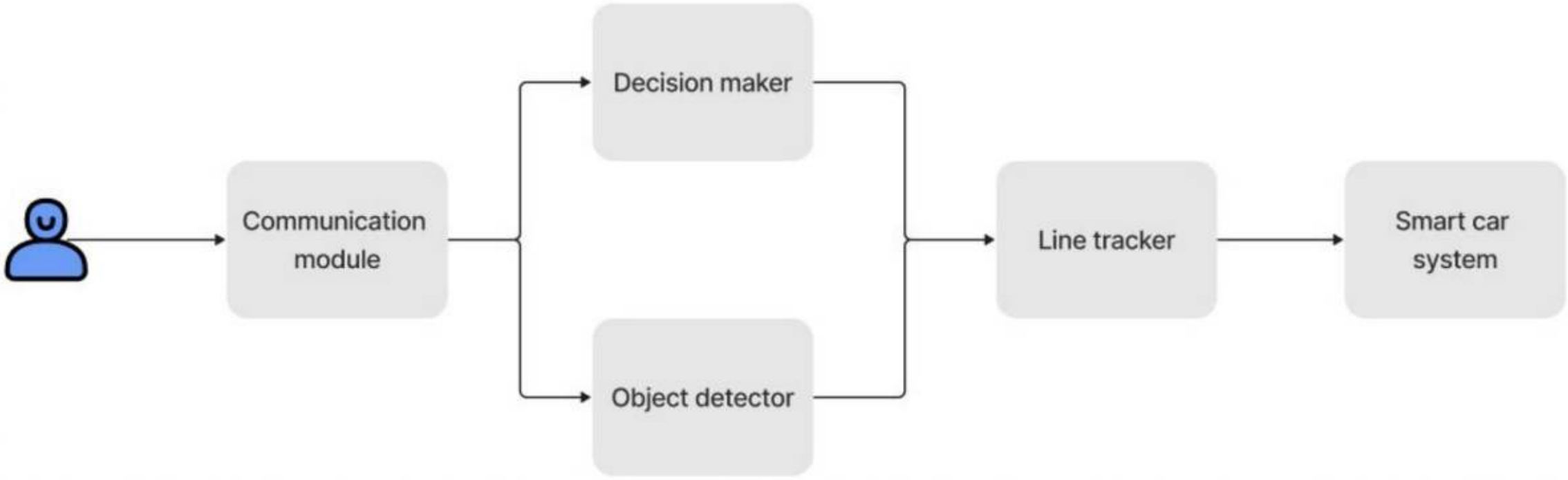


HARDWARE REQUIREMENTS:

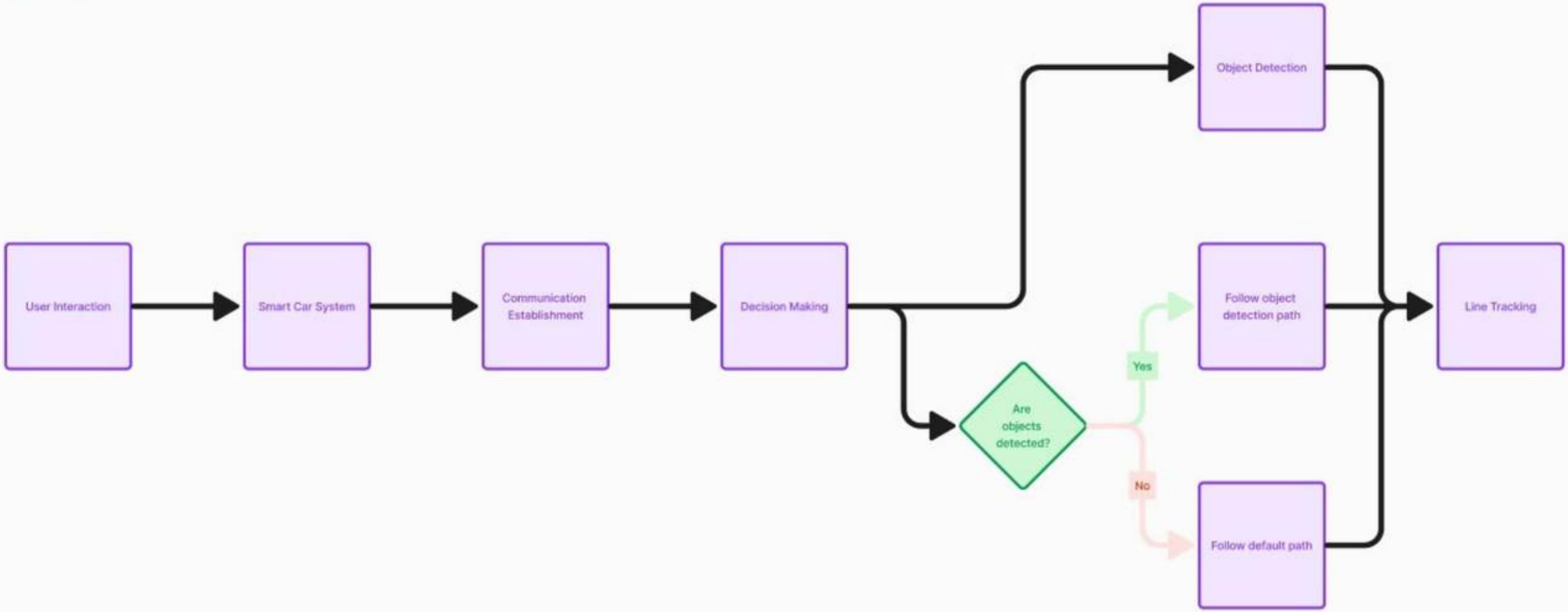


- Solderless Breadboard
- 4 wheel robot car kit
- Arduino UNO
- IR Sensor x 2
- L298 Motor Driver
- Mini Servo Motor SG90
- Ultrasonic Sensor Holder
- Ultrasonic Sensor hc-sr04
- 4Pcs Smart Robot Car Tyres Wheels
- Male to Female jumper Wires
- Male to Male jumper Wires
- Hard Jumper Wire
- On/Off Switch
- 18650 Battery Holder - 2 Cell
- 18650 Battery Cell 3.7V x 2

ARCHITECTURE:

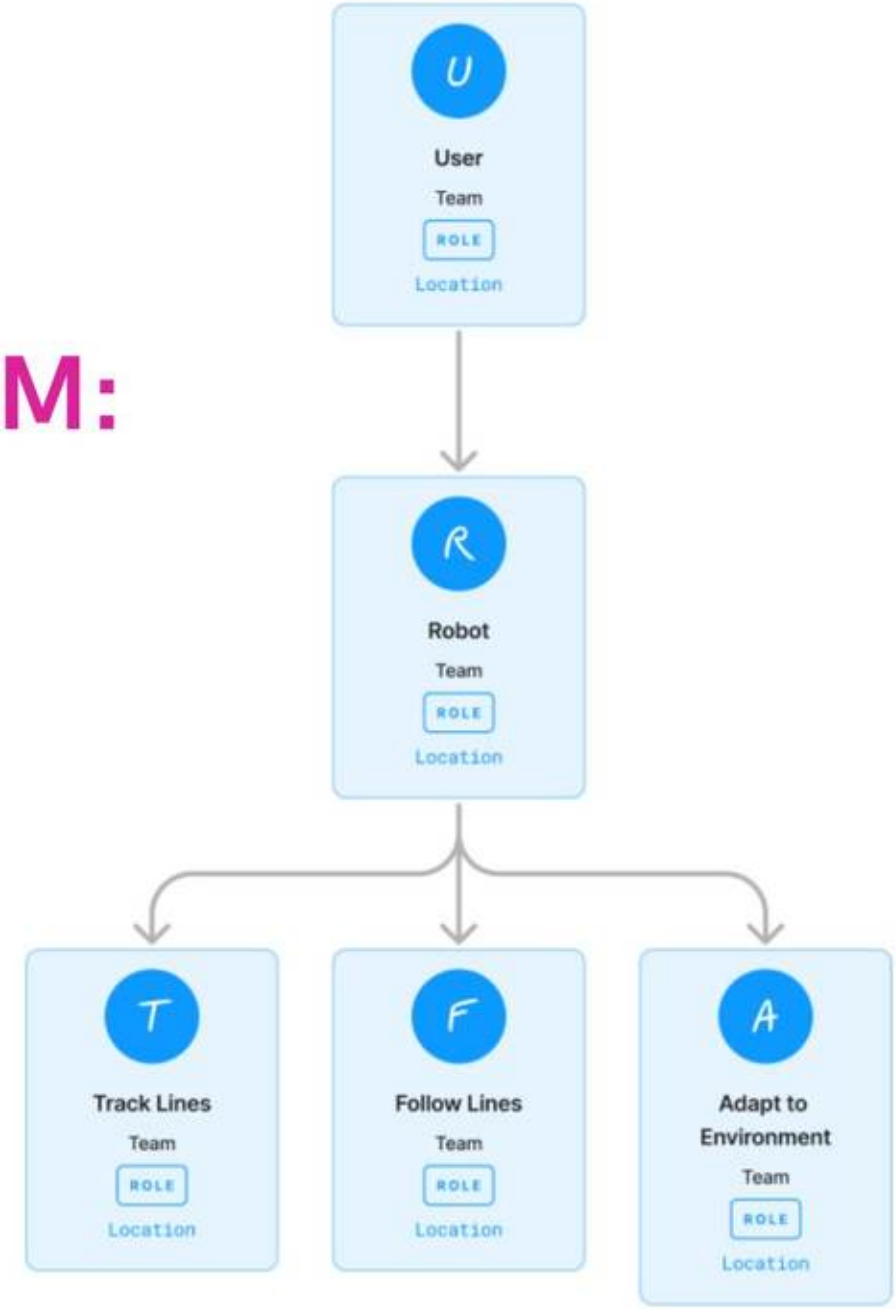


FLOW CHART :

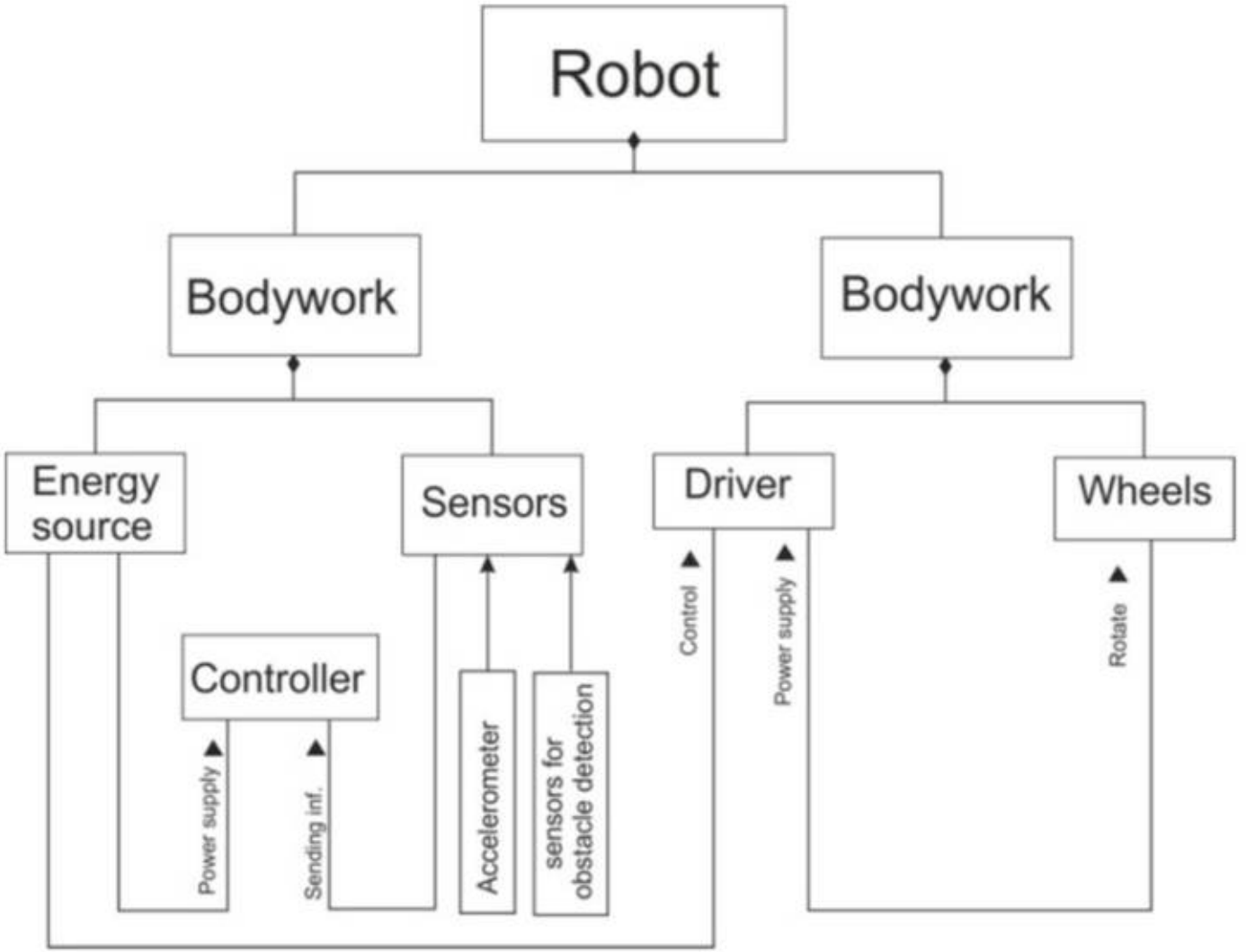

User Interaction
Smart Car System
Communication Establishment
Decision Making
Object Detection
Are objects detected?
Yes
No
Follow object detection path
Follow default path
Line Tracking

UML DIAGRAMS:

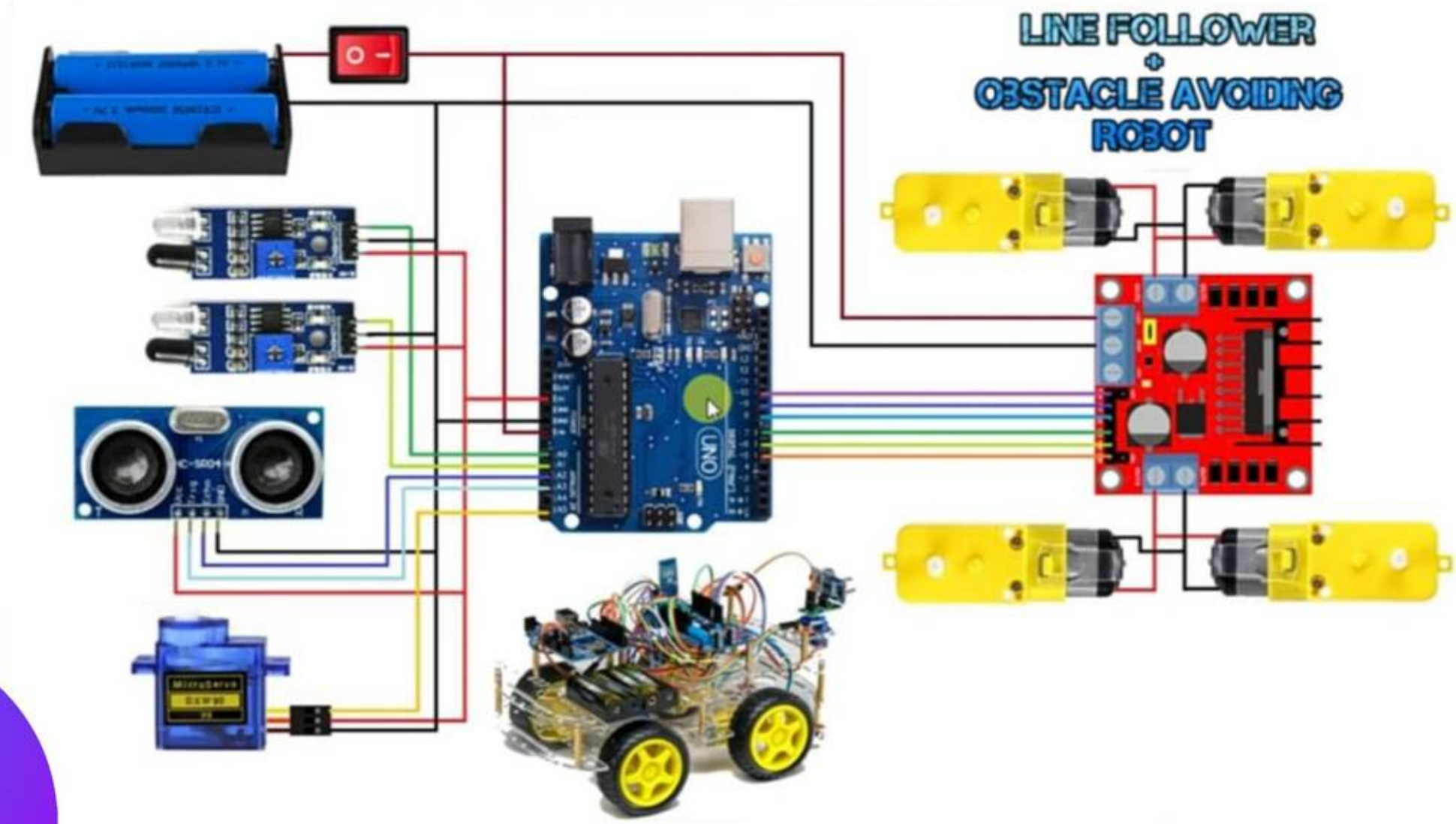
USE CASE DIAGRAM:



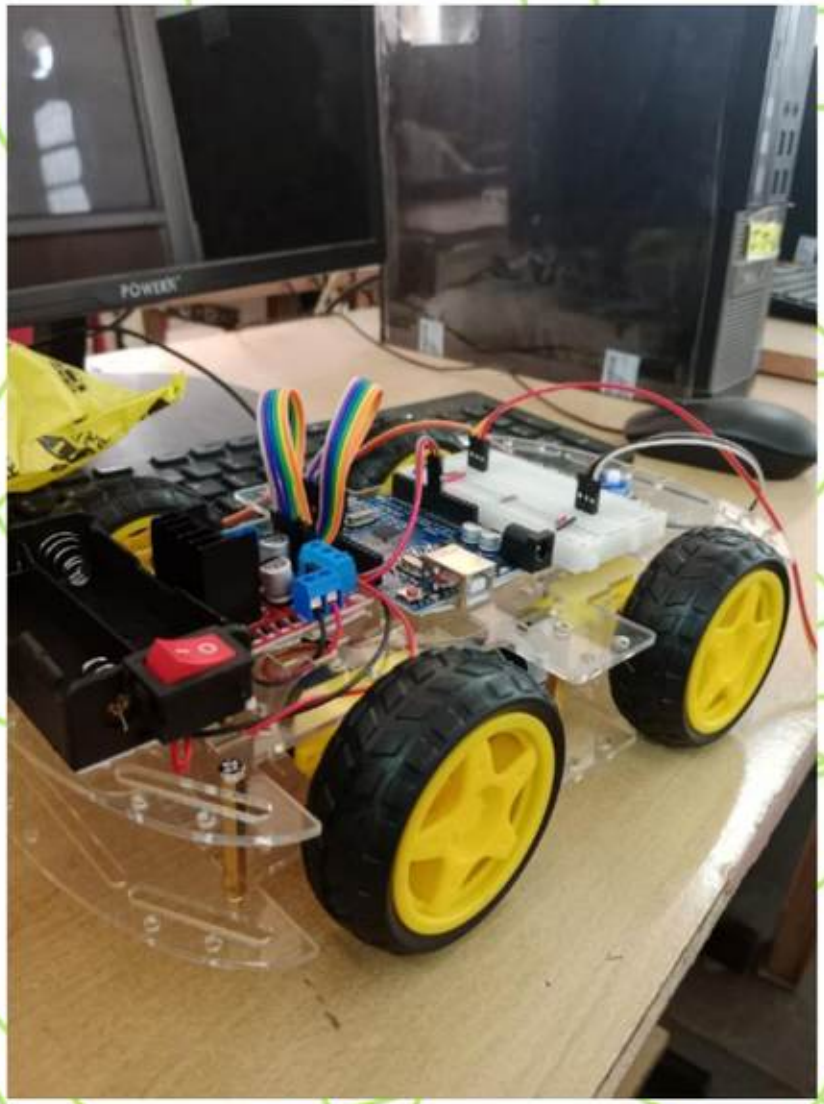
DATA FLOW DIAGRAM:

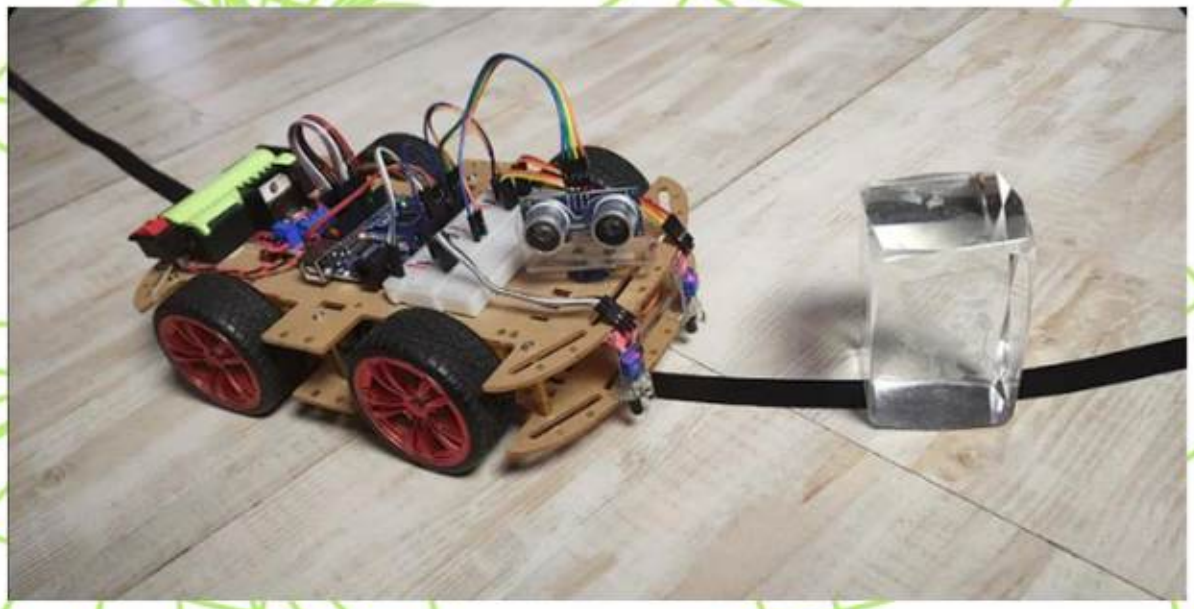


COMPONENT DIAGRAM:

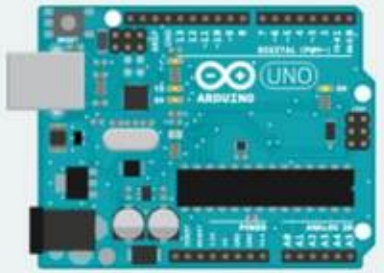


PROTOTYPE





WHAT IS ARDUINO?



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Help

HC05_LED | Arduino IDE 2.3.1

File Edit Sketch Tools Help

Select Board

HC05_LED.ino

```
#define enA 10//Enable1 L298 Pin enA
#define in1 9 //Motor1  L298 Pin in1
#define in2 8 //Motor1  L298 Pin in1
#define in3 7 //Motor2  L298 Pin in1
#define in4 6 //Motor2  L298 Pin in1
#define enB 5 //Enable2 L298 Pin enB

#define L_S A0 //ir sensor Left
#define R_S A1 //ir sensor Right

#define echo A2    //Echo pin
#define trigger A3 //Trigger pin

#define servo A5

int Set=15;
int distance_L, distance_F, distance_R;

void setup(){ // put your setup code here, to run once

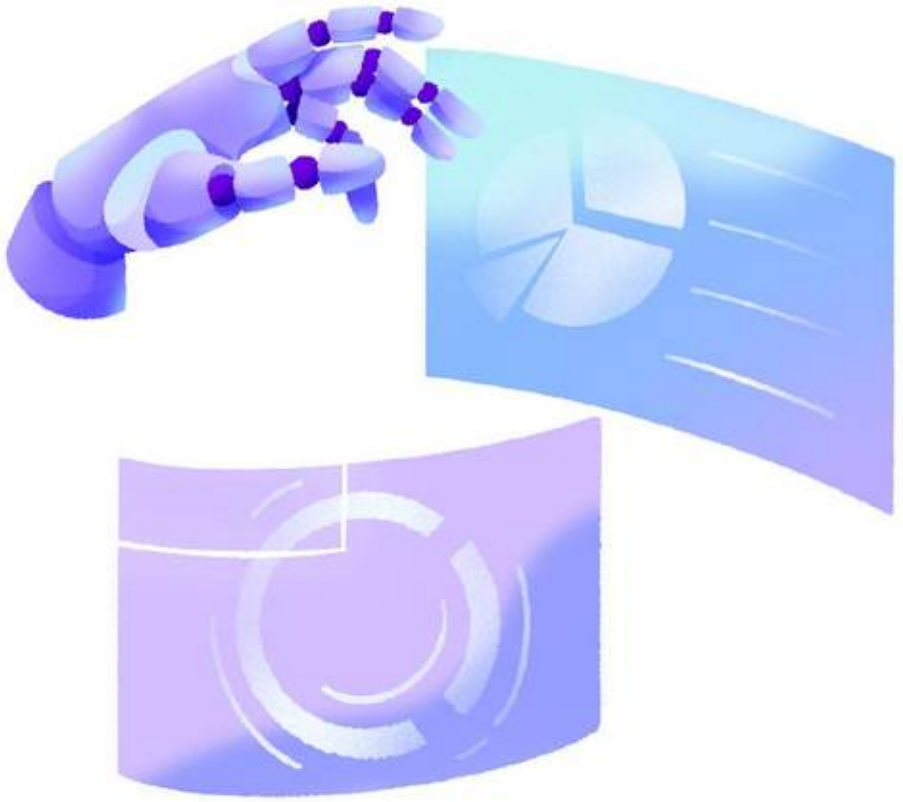
Serial.begin(9600); // start serial communication at 9600bps

pinMode(R_S, INPUT); // declare if sensor as input
pinMode(L_S, INPUT); // declare ir sensor as input

pinMode(echo, INPUT );// declare ultrasonic sensor Echo pin as input
pinMode(trigger, OUTPUT); // declare ultrasonic sensor Trigger pin as Output

pinMode(enA, OUTPUT); // declare as output for L298 Pin enA
pinMode(in1, OUTPUT); // declare as output for L298 Pin in1
pinMode(in2, OUTPUT); // declare as output for L298 Pin in2
pinMode(in3, OUTPUT); // declare as output for L298 Pin in3
pinMode(in4, OUTPUT); // declare as output for L298 Pin in4
```

ALGORITHM:



- Initialization
- Line Tracking
- Object Detection
- Sensor Fusion
- Decision-Making
- IoT Communication
- Remote Monitoring and Control
- Dynamic Path Planning
- Conversion to Fiat Currency (Optional)
- End-of-Task Handling
- Error Handling
- Shutdown and Power Management

CONCLUSION:

In summary, our IoT-integrated black line follower robot redefines precision, adaptability, and safety. With a cost-effective design and global accessibility, it sets new efficiency standards. The seamless integration of computer vision, sensor fusion, and IoT not only overcomes deficiencies but also positions the system for versatile applications. This transformative project marks a pivotal moment in robotics, contributing to IoT-enabled advancements and promising a future of sophistication and adaptability.





FUTURE ENHANCEMENT:

>INTEGRATION OF CAMERA

>IMMEDIATE RESPONSE VIA MAIL

>USER INTERFACE TO OPERATE IT

>IMPROVING EFFICIENCY ETC

THANK'S FOR WATCHING

